1. Aspect 4: Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation

- **Analysis**: The problem again relies on the payment rules shown in the poster image (e.g., commission per newspaper, fixed base salary plus commission, tiered rates, etc.). Students must correctly decode the image to establish the functional relationship between sales quantity and earnings. This is a prerequisite for solving the problem and the first step in modelling.

2. Aspect 2: Identify the key variables in a model

- **Analysis**: The key variables in this problem are "weekly earnings" (given as 74 zeds) and "weekly sales quantity" (unknown). Unlike the previous question, here earnings are the input and sales quantity is the output. Additionally, implicit variables such as the fixed component and unit commission in the payment rules must be considered. Students need to clarify the dependency among these variables.

3. Aspect 9: Translate a problem into a standard mathematical representation or algorithm

- **Analysis**: This problem requires applying the payment rules in reverse. If the forward model is `Earnings = Fixed part + Unit commission × Quantity` (or another form), solving for quantity requires solving an equation or performing an inverse operation. For example, if the rule is `Earnings = a + b × x`, then `x = (Earnings − a) / b`. Transforming the forward model into a reverse algorithm is a core mathematical step.

4. Aspect 6: Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables

- **Analysis**: Students must recognise the functional relationship between earnings and sales quantity and solve for quantity given earnings. This involves understanding monotonicity (earnings typically increase with sales) and, in the case of piecewise functions, the possibility of multiple solutions or interval determination. Real-world payment rules may include tiered commissions, requiring a piecewise approach when solving backwards.

5. Aspect 5: Recognize mathematical structure (including regularities, relationships, and patterns) in problems or situations

- **Analysis**: The payment rules often have a linear or piecewise linear structure. Students need to identify this structure and understand that its inverse operation follows the corresponding mathematical pattern. For a linear rule, the inverse is also linear; for a piecewise function, one must determine which interval the earnings fall into and then solve the equation for that interval.

6. Aspect 8: Recognize aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures

- **Analysis**: This problem is of the type "given total amount and unit rate (or value of a linear function), find the quantity", corresponding to solving a linear equation or piecewise equation in algebra. Students need to connect it to prior experience (e.g., "total price ÷ unit price = quantity" or "solving an equation") to select the correct inverse operation.

7. Aspect 7: Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

- **Analysis**: The problem assumes Christine's earnings are exactly 74 zeds, and the payment rules remain unchanged throughout the week, ignoring real-world factors such as returns, taxes, etc. It may also implicitly assume that the number of newspapers sold is an integer. Students must ignore non-essential details and simplify the problem into a purely mathematical solution.

8. Aspect 3: Select a representation appropriate to the problem context

- **Analysis**: For reverse solving, students can choose an algebraic equation, a graph (finding the corresponding point), or a table (interpolation). Since earnings are given, an algebraic equation is usually the most direct; if the rules are complex, a graphical approach can help visually determine the correct interval. Choosing an appropriate representation aids accurate solving.

9. Aspect 10: Use mathematical tools (using appropriate variables, symbols, diagrams) to describe the mathematical structures and/or relationships in a problem

- **Analysis**: Students need to assign an unknown (e.g., let sales quantity be `x`), write an equation based on the poster's rules (e.g., `74 = a + b·x` or piecewise conditions), and then solve it. This involves symbolic expression and algebraic manipulation, which is a direct application of mathematical tools.

10. Aspect 1: Select a mathematical description or a representation that describes a problem

- **Analysis**: Students must choose from models such as direct proportion, linear function, or piecewise function the one that best matches the poster's rules. For a reverse problem, the chosen model must be invertible (i.e., the function must be one-to-one). For example, if the payment rule is a fixed base salary plus commission, the linear function is invertible; if there are tiers and the earnings fall into an overlapping interval (rare), additional assumptions may be needed.

11. Aspect 12: Identify the constraints, assumptions simplifications in a mathematical model

- **Analysis**: Reverse solving relies on assumptions such as the model being invertible (e.g., strictly monotonic). Additionally, implicit assumptions include non-negative integer sales and no upper bound on the payment rules. Students need to be aware of these constraints; otherwise, they might obtain non-integer or negative results, in which case they should check or round appropriately.

12. Aspect 11 (duplicate of Aspect 10, omitted)

Summary: This problem primarily tests students' ability to extract the payment rules from the poster image (Aspect 4), build a forward model, and then solve for sales quantity in reverse (Aspect 9). Compared to the previous question, this one places significantly more emphasis on **reverse modelling and solving equations**, making **Aspect 9 (translating to a standard algorithm)** much more prominent. At the same time, identifying variables (Aspect 2) and decoding the image (Aspect 4) remain foundational. Overall, this problem strongly aligns with the "model usage and inversion" phase of mathematical modelling.

1. 方面4：阅读、解码并理解陈述、问题、任务、对象或图像，以创建情境模型

- **分析**：题目再次依赖海报图像中的支付规则（例如，每份报纸的提成、固定底薪加提成、阶梯价格等）。学生必须正确解码图像信息，建立从销量到收入的函数关系。这是解题的前提，也是建模的第一步。

2. 方面2：识别模型中的关键变量

- **分析**：本题的关键变量是“每周收入”（已知为74 zeds）和“每周销售数量”（未知）。与前一题不同，这里收入是输入，销量是输出。此外，支付规则中的固定部分、单位提成等也是隐含变量。学生需要明确这些变量之间的依赖关系。

3. 方面9：将问题转化为标准的数学表示或算法

- **分析**：本题需要逆向运用支付规则。若正向模型为 `收入 = 固定部分 + 单位提成 × 销量` （或其他形式），则求解销量就需要解方程或逆运算。例如，若规则为 `收入 = a + b × x`，则 `x = (收入 - a) / b`。这种从正向模型到逆向算法的转化是核心数学步骤。

4. 方面6：识别并描述现实问题情境中的数学方面，包括确定重要变量

- **分析**：学生需认识到收入与销量之间的函数关系，并根据已知收入反求销量。这涉及对函数单调性（通常收入随销量增加而增加）以及分段函数可能带来的多解或区间判断的理解。现实情境中的支付规则可能包含阶梯提成，此时反求销量需要分段讨论。

5. 方面5：识别问题或情境中的数学结构（包括规律、关系和模式）

- **分析**：支付规则通常具有线性或分段线性结构。学生需要识别这种结构，并理解其逆运算同样遵循相应的数学模式。例如，若规则是线性的，则逆运算也是线性的；若是分段函数，则需要确定收入落在哪个区间，再求解对应区间的方程。

6. 方面8：识别问题中与已知问题或数学概念、事实、程序相对应的方面

- **分析**：该问题属于“已知总量和单价（或线性函数值），求数量”的类型，对应代数中的解一元一次方程或分段方程。学生需将其与已有经验（如“总价÷单价=数量”或“解方程”）联系起来，从而选择正确的逆向运算。

7. 方面7：简化或分解情境或问题，以使其适合数学分析

- **分析**：题目假设Christine的收入恰好为74 zeds，且支付规则在整周内不变，不考虑退货、税收等现实因素。同时，可能隐含假设销量为整数。学生需忽略非关键细节，将问题简化为纯数学求解。

8. 方面3：选择适合问题情境的表示方式

- **分析**：对于逆向求解，学生可以选择代数方程、函数图像（从图像上找对应点）或表格（插值）等方式。由于收入是给定值，代数方程通常最直接；若规则复杂，图像法可帮助直观判断所在区间。选择合适表示有助于准确求解。

9. 方面10：使用数学工具（变量、符号、图表）描述问题中的数学结构和/或关系

- **分析**：学生需设未知数（如设销售量为 `x`），并根据海报规则写出方程（如 `74 = a + b·x` 或分段条件），然后求解。这涉及符号化表达和代数运算，是数学工具的直接应用。

10. 方面1：选择描述问题的数学描述或表示

- **分析**：学生需从正比例、线性、分段函数等模型中选择最符合海报规则的那一种。对于逆向问题，选择的模型必须可逆（即函数是单射）。例如，若支付规则是固定底薪加提成，则线性函数可逆；若存在阶梯且收入落在重叠区间（罕见），则可能需额外假设。

11. 方面12：识别数学模型中的约束、假设和简化

- **分析**：逆向求解依赖于模型的可逆性假设（如单调递增）。此外，隐含假设销量为非负整数，支付规则无上限等。学生需意识到这些约束，否则可能得到非整数或负数等不合理结果，此时应检查或取整。

12. 方面11（与方面10重复，略）

总结：该题目主要考察学生从海报图像中提取支付规则（方面4），建立正向模型后，再逆向求解销量（方面9）。与前一题相比，本题更强调**逆向建模与解方程**的能力，因此**方面9（转化为标准算法）** 的相关性显著上升。同时，识别变量（方面2）和解码图像（方面4）仍然是基础。整体上，该问题高度契合数学建模中的“模型使用与反演”环节。